WEB4 · AI AGENT DAPP

WeilChain AI Agent Chatbot

An AI-powered decentralized chatbot built on WeilChain, leveraging MCP servers and on-chain LLM execution.

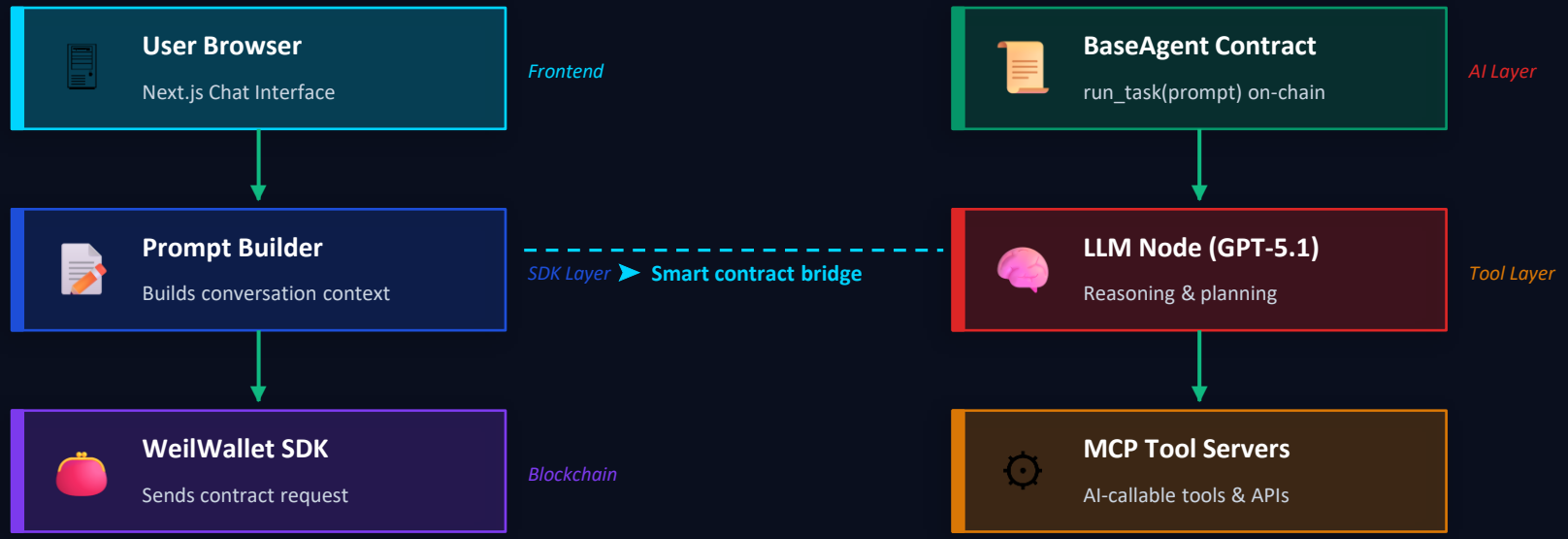
MCP Tools
Smart Contract APIs

Chat Interface
Next.js + Tailwind

WeilWallet
Blockchain SDK

System Architecture

Web4 Agent Architecture — blockchain orchestrates AI execution and tool access



Model Context Protocol (MCP)

A standardized interface enabling AI models to call external tools as smart contract applets

MCP Server = Smart Contract

Each MCP server is a smart contract applet. Functions become AI-callable tools described in structured JSON.

Tool Invocation

The LLM reasons and decides when to call tools like add(x,y) or swap_tokens(). No hardcoded logic.

Agentic Loop

Tool results return to the LLM, which synthesizes a final answer — enabling multi-step reasoning.

Example Reasoning Flow

User asks
"What is 7 × 8?"

→

LLM decides
→ call multiply(7, 8)

→

MCP returns
56

→

LLM responds
"The result is 56."

Key Features & Components

Modular, decentralized and composable AI architecture

AI Agent Smart Contract

BaseAgent + BaseAgentHelper execute AI tasks on-chain. Smart contract orchestrates AI reasoning.

MCP Tool Integration

AI model invokes tools via MCP servers. Extendable with arithmetic, DeFi, DAO, analytics tools.

Chat Interface

Built with Next.js App Router + Tailwind CSS. Components: ChatWindow, MessageInput, ConnectWallet.

Context-Aware Prompts

Frontend stores conversation history and dynamically builds structured prompts — no on-chain storage.

Minimal On-Chain State

No chat history stored on chain → faster execution, lower gas cost, scalable throughput.

Modular Architecture

Each MCP server can be independently deployed and swapped. New tools = new capabilities instantly.

Interaction Flow & Prompt Construction

10-step user journey from natural language to AI response

1. User sends a message in Chat UI
2. Frontend stores conversation history locally
3. Prompt Builder formats structured prompt
4. Frontend calls BaseAgent smart contract
5. Contract forwards request to BaseAgentHelper
6. Helper invokes LLM model on-chain
7. LLM may call MCP tool servers if needed
8. MCP tools execute their functions
9. Results return to LLM for synthesis
10. Final response displayed in Chat UI

Future Vision & Web4 Paradigm

Web4 Architecture Layers

Frontend
User interaction & prompt building

Blockchain
Agent execution & orchestration

AI Model
Reasoning, planning & decisions

MCP Servers
External tools & blockchain actions

DeFi Copilot

Natural language → blockchain transactions. Swap, stake, and manage assets via chat.

DAO Governance

AI assistants interact with governance contracts for proposals, voting, and delegation.

Multi-Agent Systems

Multiple AI agents coordinating on-chain through smart contract message passing.

Autonomous Trading

Agents executing DeFi strategies triggered by on-chain events and price feeds.

One Sentence Summary

This project demonstrates how AI agents on WeilChain can reason with LLMs and interact with decentralized tools through MCP servers, creating the foundation for programmable Web4 intelligence.